Mean and Variance in Time Series

To understand **Stationarity**, you must know **Mean** and **Variance**.

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1) Mean (Average)

Definition

The **Mean** is the average value of all observations.

Formula

Formula

$\bar{x} = \frac{x_1+x_2+x_3+\cdots+x_n}{n}$

Where:

- x = observations
- n = number of observations

Example

Daily sales:

Day	Sales
Mon	100
Tue	120
Wed	140
Thu	160
Fri	180

Calculation

Mean = (100 + 120 + 140 + 160 + 180) / 5
Mean = 700 / 5
Mean = 140

✅ Average sales = **140**

Real-Time Example of Mean

Suppose a student gets:

Subject	Mark
Maths	80
Science	90
English	70

Mean = (80 + 90 + 70)/3
Mean = 240/3 = 80

✅ Average mark = **80**

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2) Variance

Definition

Variance measures how far the data points are spread from the mean.

- Small Variance → Data points are close to mean.
- Large Variance → Data points are far from mean.

Formula

Variance=∑(xi−x¯)2nVariance=\frac{\sum (x_i-\bar{x})^2}{n}Variance=n∑(xi−x¯)2

Example 1 (Low Variance)

Data:

98, 100, 102

Step 1: Mean

(98+100+102)/3 = 100

Step 2: Difference from Mean

Value	Mean	Difference
98	100	-2
100	100	0
102	100	+2

Step 3: Square Differences

Difference	Square
-2	4
0	0
2	4

Variance = (4+0+4)/3
Variance = 2.67

✅ Small variance

The values are very close together.

Example 2 (High Variance)

Data:

20, 100, 180

Step 1: Mean

(20+100+180)/3 = 100

Step 2

Value	Mean	Difference
20	100	-80
100	100	0
180	100	+80

Step 3

Difference	Square
-80	6400
0	0
80	6400

Variance = (6400+0+6400)/3
Variance = 4266.67

❌ Very large variance

Values are spread far apart.

Why Mean and Variance Matter in Time Series?

Stationary Series

Year	Sales
1	100
2	102
3	98
4	101
5	99

Mean ≈ 100

Variance ≈ Constant

✅ Stationary

Non-Stationary Series

Year	Sales
1	100
2	200
3	300
4	400
5	500

Mean keeps changing over time.

❌ Non-Stationary

Easy Real-Life Analogy

Mean = Average Salary

Suppose 5 employees earn:

25000, 26000, 25500, 24500, 25000

Mean salary ≈ ₹25,200

Variance = Salary Difference

If salaries are:

25000, 26000, 25500, 24500, 25000

Variance is small ✅

If salaries are:

10000, 25000, 50000, 75000, 100000

Variance is huge ❌

Mean: Average value of the observations.

Variance: Measure of how much the observations vary from the mean.

For a **stationary time series**, both the **mean and variance should remain approximately constant over time.**

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